

MEDICAL BIOCHEMISTRY MASTER PRACTICE BANK

*100 Multiple-Choice Questions on Enzyme Kinetics, Mechanisms, & Clinical Diagnostics
Grounded in Harvey & Ferrier's Biochemistry & Lieberman's Marks' Basic Medical Biochemistry*

TOPIC 1: Definition & Chemical Nature of Enzymes

Question 1

What is the primary chemical nature of virtually all biological enzymes, with the exception of catalytic RNAs?

- A. Linear unbranched polysaccharides
- B. Globular proteins composed of L-amino acids
- C. Double-stranded deoxyribonucleic acid polymers
- D. Esterified triacylglycerol lipid complexes

Correct Answer: B. Globular proteins composed of L-amino acids

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Polysaccharides like glycogen serve structural or energy storage roles; they lack catalytic active sites.

[B] (**CORRECT**): Enzymes are globular proteins folded into precise 3D tertiary/quaternary conformations creating specific catalytic clefts.

[C] (**INCORRECT**): DNA stores genetic information; while catalytic RNAs (ribozymes) exist, native DNA lacks intrinsic catalytic activity.

[D] (**INCORRECT**): Triacylglycerols are hydrophobic storage lipids without active site catalytic capability.

Question 2

An enzyme accelerates a chemical reaction rate by altering which of the following thermodynamic parameters?

- A. Standard free energy change (ΔG°)
- B. Activation energy barrier ($\Delta G^\ddagger$) of the transition state
- C. Equilibrium constant (K_{eq}) of the reaction
- D. Total enthalpy (ΔH) of initial reactants and final products

Correct Answer: B. Activation energy barrier ($\Delta G^\ddagger$) of the transition state

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Enzymes do not alter ΔG° ; the net free energy difference between initial reactants and final products is unchanged.

[B] (**CORRECT**): Enzymes lower $\Delta G^\ddagger$ by stabilizing the transition state, allowing more molecules to react per unit time.

[C] (**INCORRECT**): Enzymes accelerate the rate of reaching equilibrium but do not alter the final equilibrium ratio (K_{eq}).

[D] (**INCORRECT**): Enthalpy ΔH is a thermodynamic state function of reactants and products and is unchanged by catalysts.

Question 3

Which statement correctly describes the effect of an enzyme on a reversible reaction at chemical equilibrium?

- A. Increases the forward reaction rate while decreasing the reverse rate
- B. Increases both forward and reverse rates equally without shifting K_{eq}
- C. Shifts the final equilibrium position toward product formation
- D. Inputs net free energy to increase the final product-to-substrate ratio

Correct Answer: B. Increases both forward and reverse rates equally without shifting K_{eq}

Detailed Choice-by-Choice Rationale:

- [A] (**INCORRECT**): Enzymes lower the activation energy barrier for both directions equally; favoring one direction violates thermodynamics.
- [B] (**CORRECT**): Because $\Delta G^\ddagger$ is lowered for both forward and reverse pathways, both rates increase by the exact same factor.
- [C] (**INCORRECT**): The ratio [Products]/[Reactants] at equilibrium (K_{eq}) depends solely on ΔG° and remains unchanged.
- [D] (**INCORRECT**): Enzymes are catalysts; they are neither consumed nor do they provide net energy to shift chemical equilibrium.

Question 4

The ability of an enzyme to distinguish its specific substrate from close structural analogs is dictated by:

- A. Covalent peptide bonds in the primary sequence alone
- B. Precise 3D complementary geometry and non-covalent interactions at the active site
- C. Non-specific hydrophobic collapse of the entire protein core
- D. The presence of an N-terminal signal peptide sequence

Correct Answer: B. Precise 3D complementary geometry and non-covalent interactions at the active site

Detailed Choice-by-Choice Rationale:

- [A] (**INCORRECT**): Primary sequence determines linear amino acid order, but 3D folding is mandatory for substrate binding pockets.
- [B] (**CORRECT**): Complementary spatial shape, hydrogen bonding, ionic charges, and van der Waals forces confer high specificity.
- [C] (**INCORRECT**): Hydrophobic core collapse stabilizes overall folding but does not dictate selective substrate recognition.
- [D] (**INCORRECT**): Signal peptides direct subcellular protein translocation and are cleaved during protein maturation.

Question 5

Catalytic RNA molecules that function as biological catalysts are termed:

- A. Zymogens
- B. Ribozymes
- C. Isozymes
- D. Abzymes

Correct Answer: B. Ribozymes

Detailed Choice-by-Choice Rationale:

- [A] (**INCORRECT**): Zymogens are inactive protein proenzymes that require proteolytic cleavage for activation.
- [B] (**CORRECT**): Ribozymes are catalytic RNA molecules (e.g., 28S rRNA peptidyl transferase, RNase P).
- [C] (**INCORRECT**): Isozymes are distinct protein variants of an enzyme that catalyze the same reaction.
- [D] (**INCORRECT**): Abzymes are catalytic antibodies engineered to bind transition state analogs.

Question 6

The maximum rate acceleration achieved by typical biological enzymes compared to uncatalyzed reactions ranges between:

- A. 2-fold to 5-fold
- B. 10-fold to 100-fold
- C. 10^3 -fold to 10^5 -fold
- D. 10^6 -fold to 10^{17} -fold

Correct Answer: D. 10^6 -fold to 10^{17} -fold

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): 2- to 5-fold accelerations reflect mild physical heating, not enzymatic catalysis.

[B] (**INCORRECT**): Simple inorganic chemical catalysts achieve 10^2 -fold accelerations.

[C] (**INCORRECT**): 10^3 - to 10^5 -fold increases represent weak artificial catalysts, far below biological enzymes.

[D] (**CORRECT**): Biological enzymes enhance reaction rates by factors of 10^6 to 10^{17} (e.g., Orotidine monophosphate decarboxylase).

Question 7

Complete heat denaturation inactivates an enzyme's catalytic capacity primarily because denaturation:

- A. Hydrolyzes covalent peptide bonds along the backbone
- B. Disrupts non-covalent secondary, tertiary, and quaternary structures required for active site geometry
- C. Chemically alters the covalent structure of the substrate molecule
- D. Converts the enzyme into a competitive inhibitor

Correct Answer: B. Disrupts non-covalent secondary, tertiary, and quaternary structures required for active site geometry

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Heat denaturation unfolds 3D protein structure without breaking covalent peptide bonds.

[B] (**CORRECT**): Loss of hydrogen bonds, ionic links, and hydrophobic packing destroys the specific catalytic active site.

[C] (**INCORRECT**): Heat affects the protein's native state; substrate chemical bonds remain intact prior to binding.

[D] (**INCORRECT**): Denatured proteins aggregate and lose binding capability altogether rather than acting as competitive inhibitors.

Question 8

What distinguishes absolute substrate specificity from group specificity in enzyme action?

- A. Absolute specificity acts on only one substrate, while group specificity acts on structurally related analogs
- B. Absolute specificity involves irreversible binding; group specificity is always reversible
- C. Group specificity applies only to ribozymes; absolute specificity applies to protein enzymes
- D. Absolute specificity requires metal cofactors; group specificity requires organic coenzymes

Correct Answer: A. Absolute specificity acts on only one substrate, while group specificity acts on structurally related analogs

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Urease shows absolute specificity (only urea), whereas hexokinase shows group specificity (glucose, fructose, mannose).

[B] (**INCORRECT**): Specificity defines substrate selectivity, not the reversibility of enzyme-substrate binding.

[C] (**INCORRECT**): Both protein enzymes and ribozymes exhibit varying degrees of substrate specificity.

[D] (**INCORRECT**): Cofactor requirements are independent of the breadth of substrate specificity.

Question 9

According to transition state theory, the catalytic efficiency of an enzyme relies heavily on its ability to bind:

- A. The ground-state substrate much more tightly than the transition state
- B. The high-energy transition state complex with higher affinity than ground-state substrate or product
- C. The reaction product with infinite affinity to prevent reverse reaction
- D. Competitive inhibitors exclusively during catalytic turnover

Correct Answer: B. The high-energy transition state complex with higher affinity than ground-state substrate or product

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Tightly binding ground-state substrate creates a thermodynamic energy well, increasing activation energy.

[B] (**CORRECT**): Preferential binding and stabilization of the high-energy transition state significantly lowers $\Delta G^\ddagger$.

[C] (**INCORRECT**): Binding product with infinite affinity causes product inhibition and halts enzyme turnover.

[D] (**INCORRECT**): Inhibitor binding blocks catalytic activity rather than facilitating reaction rate acceleration.

Question 10

Which structural level of a protein enzyme contains the exact linear sequence of amino acids linked by peptide bonds?

- A. Primary structure
- B. Secondary structure
- C. Tertiary structure
- D. Quaternary structure

Correct Answer: A. Primary structure

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Primary structure is the unique linear sequence of amino acids joined covalently by peptide bonds.

[B] (**INCORRECT**): Secondary structure refers to local folding patterns like α -helices and β -sheets.

[C] (**INCORRECT**): Tertiary structure represents the overall 3D folding of a single polypeptide chain.

[D] (**INCORRECT**): Quaternary structure describes the spatial arrangement of multiple polypeptide subunits.

Question 11

Binding energy (ΔG_B) in enzymatic catalysis is defined as the free energy:

- A. Released during peptide bond cleavage
- B. Derived from non-covalent interactions between enzyme and transition state substrate
- C. Consumed during ATP hydrolysis to drive reactions
- D. Difference between initial reactants and final products (ΔG°)

Correct Answer: B. Derived from non-covalent interactions between enzyme and transition state substrate

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Peptide bond cleavage is a hydrolytic reaction, not binding energy.

[B] (**CORRECT**): Multiple weak non-covalent interactions formed in the transition state release binding energy that offsets $\Delta G^\ddagger$.

[C] (**INCORRECT**): ATP hydrolysis provides energy for endergonic coupling, distinct from intrinsic catalytic binding energy.

[D] (**INCORRECT**): ΔG° is the overall thermodynamic reaction energy change, independent of binding energy.

Question 12

Which of the following statements regarding ribozymes is CORRECT?

- A. They are translated on ribosomes attached to the rough endoplasmic reticulum
- B. They catalyze specific RNA splicing and peptide bond formation (peptidyl transferase)
- C. They are completely immune to thermal denaturation
- D. They require lipid cofactors for all catalytic events

Correct Answer: B. They catalyze specific RNA splicing and peptide bond formation (peptidyl transferase)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Ribozymes are non-coding RNA transcripts synthesized by RNA polymerases, not translated proteins.

[B] (**CORRECT**): 28S/23S rRNA acts as peptidyl transferase during translation; spliceosomal snRNAs catalyze intron splicing.

[C] (**INCORRECT**): RNA tertiary structures unfold at elevated temperatures, resulting in thermal inactivation.

[D] (**INCORRECT**): Ribozymes rely on divalent metal cations (e.g., Mg^{2+}), not lipid cofactors.

TOPIC 2: Sites of Enzyme Synthesis & Subcellular Compartmentalization

Question 13

Where are enzymes destined for the cytosolic matrix (e.g., glycolytic enzymes) translated?

- A. Free polyribosomes in the cytosol
- B. Ribosomes bound to the outer membrane of the Rough Endoplasmic Reticulum
- C. Matrix ribosomes inside the mitochondria exclusively
- D. The inner lumen of the Golgi apparatus

Correct Answer: A. Free polyribosomes in the cytosol

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Proteins remaining in the cytosol (glycolysis, pentose phosphate pathway) are translated on free cytosolic ribosomes.

[B] (**INCORRECT**): RER-bound ribosomes synthesize secretory, lysosomal, and plasma membrane proteins.

[C] (**INCORRECT**): Mitochondrial ribosomes synthesize only 13 hydrophobic subunits of the electron transport chain encoded by mtDNA.

[D] (**INCORRECT**): The Golgi apparatus modifies and packages proteins but does not carry out protein translation.

Question 14

Secretory digestive zymogens (e.g., trypsinogen, pepsinogen) are synthesized by ribosomes attached to the:

- A. Outer nuclear envelope exclusively
- B. Rough Endoplasmic Reticulum (RER)
- C. Smooth Endoplasmic Reticulum (SER)
- D. Inner mitochondrial membrane

Correct Answer: B. Rough Endoplasmic Reticulum (RER)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Nuclear envelope ribosomes synthesize some local proteins but not major digestive zymogens.

[B] (**CORRECT**): Secretory proteins contain N-terminal signal sequences that dock ribosomes onto RER translocons for luminal insertion.

[C] (**INCORRECT**): SER is dedicated to lipid/steroid biosynthesis and xenobiotic detoxification, lacking protein-synthesizing ribosomes.

[D] (**INCORRECT**): Mitochondrial membrane facilitates oxidative phosphorylation, not extracellular digestive enzyme secretion.

Question 15

Targeting of acid hydrolase enzymes to lysosomes requires post-translational addition of which tag in the Golgi apparatus?

- A. Ubiquitin polymer
- B. Mannose-6-phosphate (M6P)
- C. GPI anchor
- D. Palmitoyl group

Correct Answer: B. Mannose-6-phosphate (M6P)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Ubiquitination targets cytosolic proteins to the 26S proteasome for degradation.

[B] (**CORRECT**): GlcNAc phosphotransferase adds M6P tags to N-linked oligosaccharides, directing enzymes to lysosomes (defective in I-cell disease).

[C] (**INCORRECT**): GPI anchors attach proteins to the outer leaflet of the plasma membrane.

[D] (**INCORRECT**): Palmitoylation anchors cytosolic proteins to intracellular membranes.

Question 16

Which of the following metabolic pathways contains enzymes located exclusively within the mitochondrial matrix?

- A. Glycolysis
- B. Tricarboxylic Acid (TCA) Cycle (except succinate dehydrogenase)
- C. Pentose Phosphate Pathway
- D. De Novo Fatty Acid Synthesis

Correct Answer: B. Tricarboxylic Acid (TCA) Cycle (except succinate dehydrogenase)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Glycolysis enzymes are located in the cytosol.

[B] (**CORRECT**): TCA cycle enzymes reside in the mitochondrial matrix (succinate dehydrogenase is bound to inner membrane).

[C] (**INCORRECT**): Pentose phosphate pathway enzymes are cytosolic.

[D] (**INCORRECT**): De novo fatty acid synthesis is catalyzed by cytosolic Fatty Acid Synthase.

Question 17

Which organelle houses catalase and urate oxidase for peroxisomal H₂O₂ detoxification and long-chain fatty acid oxidation?

- A. Lysosome
- B. Peroxisome
- C. Golgi complex
- D. Nucleolus

Correct Answer: B. Peroxisome

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Lysosomes contain acid hydrolases (pH ~4.5–5.0) for macromolecular degradation.

[B] (**CORRECT**): Peroxisomes contain catalase and oxidases to metabolize H₂O₂ and initiate α - and β -oxidation of very long chain fatty acids (VLCFA).

[C] (**INCORRECT**): Golgi modifies carbohydrates and sorts proteins.

[D] (**INCORRECT**): Nucleolus is the site of rRNA transcription and ribosome subunit assembly.

Question 18

Which enzyme of the urea cycle is localized specifically in the mitochondrial matrix, while the remaining steps occur in cytosol?

- A. Arginase
- B. Carbamoyl phosphate synthetase I (CPS-I)
- C. Argininosuccinate lyase
- D. Argininosuccinate synthetase

Correct Answer: B. Carbamoyl phosphate synthetase I (CPS-I)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Arginase is a cytosolic enzyme cleaving arginine into urea and ornithine.

[B] (**CORRECT**): CPS-I (and Ornithine transcarbamylase) reside in the mitochondrial matrix; remaining steps occur in the cytosol.

[C] (**INCORRECT**): Argininosuccinate lyase operates in the cytosol.

[D] (**INCORRECT**): Argininosuccinate synthetase is cytosolic.

Question 19

Glucose-6-phosphatase, the final enzyme of gluconeogenesis and glycogenolysis, is anchored to the membrane of the:

- A. Mitochondrial inner membrane
- B. Endoplasmic Reticulum (ER)
- C. Plasma membrane outer surface
- D. Peroxisomal matrix

Correct Answer: B. Endoplasmic Reticulum (ER)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Mitochondrial inner membrane houses electron transport complexes, not Glucose-6-phosphatase.

[B] (**CORRECT**): Glucose-6-phosphatase is embedded in the ER membrane with its catalytic site facing the ER lumen (gluconeogenic tissues).

[C] (**INCORRECT**): Plasma membrane contains GLUT transporters, not G6Pase.

[D] (**INCORRECT**): Peroxisomes do not participate in glucose dephosphorylation.

Question 20

Nuclear-encoded mitochondrial enzymes are imported into mitochondria via which protein translocation complexes?

- A. SRP and Sec61 complex
- B. TOM and TIM complexes
- C. Clathrin and AP-2 complexes
- D. COP-I and COP-II coatamers

Correct Answer: B. TOM and TIM complexes

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): SRP and Sec61 direct co-translational insertion into the ER.

[B] (**CORRECT**): TOM (Translocase of Outer Membrane) and TIM (Translocase of Inner Membrane) import mitochondrial precursor proteins.

[C] (**INCORRECT**): Clathrin mediates endocytosis from the plasma membrane.

[D] (**INCORRECT**): COP-I and COP-II mediate vesicular transport between ER and Golgi.

Question 21

HMG-CoA synthase exists as two distinct tissue/compartment-specific isoenzymes. Cytosolic HMG-CoA synthase functions in:

- A. Ketogenesis
- B. Cholesterol biosynthesis
- C. Beta-oxidation of fatty acids
- D. TCA cycle regulation

Correct Answer: B. Cholesterol biosynthesis

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Mitochondrial HMG-CoA synthase catalyzes the rate-limiting step of ketogenesis in liver mitochondria.

[B] (**CORRECT**): Cytosolic HMG-CoA synthase synthesizes HMG-CoA for de novo cholesterol biosynthesis.

[C] (**INCORRECT**): Beta-oxidation is mitochondrial/peroxisomal and does not use HMG-CoA synthase.

[D] (**INCORRECT**): TCA cycle operates in mitochondrial matrix via citrate synthase, not HMG-CoA synthase.

Question 22

Pyruvate carboxylase, which converts pyruvate to oxaloacetate for gluconeogenesis, is located in the:

- A. Cytosol
- B. Mitochondrial matrix
- C. Lysosomal lumen
- D. Nucleolus

Correct Answer: B. Mitochondrial matrix

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Cytosol contains phosphoenolpyruvate carboxykinase (PEPCK in humans is both cytosolic and mitochondrial).

[B] (**CORRECT**): Pyruvate carboxylase is strictly an ATP-dependent biotin-containing mitochondrial matrix enzyme.

[C] (**INCORRECT**): Lysosomes degrade macromolecules and lack gluconeogenic enzymes.

[D] (**INCORRECT**): Nucleolus handles ribosome assembly.

Question 23

The enzymes responsible for DNA replication (DNA Polymerases α , δ , ϵ) and RNA transcription reside in the:

- A. Cytosol
- B. Nucleus
- C. Smooth Endoplasmic Reticulum
- D. Lysosomes

Correct Answer: B. Nucleus

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Cytosol translates mRNA but does not replicate genomic DNA.

[B] (**CORRECT**): Genomic DNA replication, repair, and RNA transcription occur inside the nucleus.

[C] (**INCORRECT**): SER synthesizes lipids and detoxifies xenobiotics.

[D] (**INCORRECT**): Lysosomes hydrolyze cellular debris.

Question 24

Acid phosphatases and cathepsin proteases are characteristic marker enzymes for which organelle?

- A. Peroxisome
- B. Lysosome
- C. Mitochondria
- D. Golgi apparatus

Correct Answer: B. Lysosome

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Peroxisomes are characterized by catalase and D-amino acid oxidase.

[B] (**CORRECT**): Lysosomes contain over 50 acidic hydrolases, including acid phosphatase, cathepsins, and glycosidases.

[C] (**INCORRECT**): Mitochondria are marked by citrate synthase and cytochrome c oxidase.

[D] (**INCORRECT**): Golgi is marked by galactosyltransferase.

TOPIC 3: Organic Catalysts, Catalytic Sites, ES Complex & Cofactors

Question 25

An enzyme devoid of its required cofactor/coenzyme is catalytically inactive and is termed an:

- A. Holoenzyme
- B. Apoenzyme
- C. Proenzyme
- D. Metalloenzyme

Correct Answer: B. Apoenzyme

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Holoenzyme is the active, intact enzyme-cofactor complex.

[B] (**CORRECT**): Apoenzyme refers to the protein component alone, which is catalytically inactive without its cofactor.

[C] (**INCORRECT**): Proenzyme (zymogen) is an inactive precursor requiring proteolytic cleavage.

[D] (**INCORRECT**): Metalloenzyme is an enzyme containing tightly bound metal ions.

Question 26

The Induced Fit Model of enzyme-substrate binding proposed by Koshland asserts that:

- A. The active site is a completely rigid template matching the substrate ground state perfectly
- B. Substrate binding induces dynamic conformational shifts in the active site to optimize transition state complementary fit
- C. Enzymes alter their primary amino acid sequence upon binding substrate
- D. Substrates bind covalently to allosteric sites before entering the active site

Correct Answer: B. Substrate binding induces dynamic conformational shifts in the active site to optimize transition state complementary fit

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Fischer's Lock and Key model proposed a rigid template, which fails to explain transition state stabilization.

[B] (**CORRECT**): Koshland's Induced Fit model highlights flexibility: substrate binding reorganizes active site residues around the transition state.

[C] (**INCORRECT**): Primary sequence is fixed by mRNA coding and does not change during substrate binding.

[D] (**INCORRECT**): Substrates bind directly to the active catalytic site, not allosteric sites.

Question 27

Which coenzyme derived from Vitamin B6 (Pyridoxine) is required for amino acid transamination reactions?

- A. Thiamine pyrophosphate (TPP)
- B. Pyridoxal phosphate (PLP)
- C. Flavin adenine dinucleotide (FAD)
- D. Nicotinamide adenine dinucleotide (NAD+)

Correct Answer: B. Pyridoxal phosphate (PLP)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): TPP (Vitamin B1) is required for oxidative decarboxylation of α -keto acids.

[B] (**CORRECT**): PLP (Vitamin B6) forms a Schiff base with amino acids, acting as an essential coenzyme for transaminases (ALT, AST).

[C] (**INCORRECT**): FAD (Vitamin B2) participates in electron transfer redox reactions.

[D] (**INCORRECT**): NAD+ (Vitamin B3) accepts hydride ions in dehydrogenase reactions.

Question 28

How does a prosthetic group differ from a cosubstrate coenzyme?

- A. Prosthetic groups are inorganic metals; cosubstrates are organic vitamins
- B. Prosthetic groups are permanently covalently or tightly bound; cosubstrates bind transiently and dissociate
- C. Cosubstrates lower activation energy; prosthetic groups alter K_{eq}
- D. Prosthetic groups function only in ribozymes

Correct Answer: B. Prosthetic groups are permanently covalently or tightly bound; cosubstrates bind transiently and dissociate

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Prosthetic groups can be organic (e.g., FAD, Heme, Biotin) or inorganic.

[B] (**CORRECT**): Prosthetic groups (FAD, Biotin, TPP) remain bound throughout catalysis; cosubstrates (NAD+, CoA) bind and release like substrates.

[C] (**INCORRECT**): Neither alters K_{eq} .

[D] (**INCORRECT**): Prosthetic groups function predominantly in protein enzymes.

Question 29

Carbonic anhydrase contains which essential metal cofactor tightly bound to three histidine residues at its active site?

- A. Iron (Fe^{2+})
- B. Zinc (Zn^{2+})
- C. Magnesium (Mg^{2+})
- D. Copper (Cu^{2+})

Correct Answer: B. Zinc (Zn^{2+})

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Iron is essential in hemoglobin, cytochromes, and catalase.

[B] (**CORRECT**): Zinc (Zn^{2+}) activates water molecules for nucleophilic attack on CO_2 in carbonic anhydrase and carboxypeptidase.

[C] (**INCORRECT**): Magnesium stabilizes ATP phosphate charges in kinase reactions.

[D] (**INCORRECT**): Copper operates in cytochrome c oxidase and superoxide dismutase.

Question 30

Thiamine Pyrophosphate (TPP), derived from Vitamin B1, serves as an essential coenzyme for which enzyme complex?

- A. Lactate Dehydrogenase
- B. Pyruvate Dehydrogenase complex
- C. Alanine Aminotransferase
- D. Acetyl-CoA Carboxylase

Correct Answer: B. Pyruvate Dehydrogenase complex

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Lactate dehydrogenase uses NAD⁺/NADH.

[B] (CORRECT): TPP is the prosthetic group for E1 (pyruvate decarboxylase subunit) of Pyruvate Dehydrogenase and α -Ketoglutarate Dehydrogenase.

[C] (INCORRECT): Alanine aminotransferase requires PLP (Vitamin B6).

[D] (INCORRECT): Acetyl-CoA carboxylase requires Biotin (Vitamin B7).

Question 31

Biotin (Vitamin B7) serves as a prosthetic group specifically for which class of enzymatic reactions?

- A. Transamination
- B. Carboxylation (CO₂ fixation)
- C. Hydrolytic cleavage
- D. Isomerization

Correct Answer: B. Carboxylation (CO₂ fixation)

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Transamination requires Pyridoxal phosphate (PLP).

[B] (CORRECT): Biotin covalently binds CO₂ to transfer it in carboxylase reactions (Pyruvate carboxylase, Acetyl-CoA carboxylase).

[C] (INCORRECT): Hydrolysis uses water and hydrolase enzymes without biotin.

[D] (INCORRECT): Isomerization rearrangements do not require biotin.

Question 32

Nicotinamide Adenine Dinucleotide (NAD⁺), derived from Niacin (Vitamin B3), participates in oxidoreductase reactions as a:

- A. Two-electron hydride ion (H⁻) acceptor
- B. Single electron transfer metal ion
- C. Covalent acyl group carrier
- D. Carbon dioxide donor

Correct Answer: A. Two-electron hydride ion (H⁻) acceptor

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): NAD⁺ accepts a hydride ion (:H⁻, carrying 2 electrons and 1 proton) to form NADH, releasing H⁺ into solution.

[B] (INCORRECT): Single electron transfers are characteristic of cytochromes (Fe³⁺/Fe²⁺) and FAD semi-quinones.

[C] (INCORRECT): Coenzyme A and Lipxamide carry acyl groups.

[D] (INCORRECT): Biotin carries CO₂.

Question 33

Which vitamin precursor is required for the synthesis of Coenzyme A (CoA-SH)?

- A. Riboflavin (Vitamin B2)
- B. Pantothenic Acid (Vitamin B5)
- C. Folic Acid (Vitamin B9)
- D. Cobalamin (Vitamin B12)

Correct Answer: B. Pantothenic Acid (Vitamin B5)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Riboflavin forms FMN and FAD.

[B] (**CORRECT**): Pantothenic acid (B5) forms the structural core of Coenzyme A and acyl carrier protein (ACP).

[C] (**INCORRECT**): Folic acid forms tetrahydrofolate (THF) for 1-carbon transfers.

[D] (**INCORRECT**): Cobalamin forms methylcobalamin and adenosylcobalamin.

Question 34

Kinase enzymes (e.g., Hexokinase, Protein Kinase A) absolute require which divalent cation to complex with ATP?

- A. Calcium (Ca^{2+})
- B. Magnesium (Mg^{2+})
- C. Copper (Cu^{2+})
- D. Manganese (Mn^{2+})

Correct Answer: B. Magnesium (Mg^{2+})

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Calcium regulates calmodulin and blood clotting factors.

[B] (**CORRECT**): Mg^{2+} neutralizes negative charges on ATP phosphate groups, forming Mg-ATP^{2-} as the true enzymatic substrate.

[C] (**INCORRECT**): Copper participates in redox electron transport.

[D] (**INCORRECT**): Manganese activates specific glycosyltransferases and arginase.

Question 35

Flavin Adenine Dinucleotide (FAD), derived from Riboflavin (Vitamin B2), is tightly bound to which TCA cycle enzyme?

- A. Citrate Synthase
- B. Succinate Dehydrogenase
- C. Malate Dehydrogenase
- D. Isocitrate Dehydrogenase

Correct Answer: B. Succinate Dehydrogenase

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Citrate synthase uses acetyl-CoA and oxaloacetate without FAD.

[B] (**CORRECT**): Succinate dehydrogenase (Complex II) contains covalently bound FAD to oxidize succinate to fumarate.

[C] (**INCORRECT**): Malate dehydrogenase uses NAD^+ .

[D] (**INCORRECT**): Isocitrate dehydrogenase uses NAD^+ or NADP^+ .

Question 36

The catalytic triad of serine proteases (e.g., Chymotrypsin) consists of which three amino acid residues?

- A. Serine, Histidine, Aspartate
- B. Cysteine, Lysine, Glutamate
- C. Threonine, Arginine, Alanine
- D. Tyrosine, Glycine, Proline

Correct Answer: A. Serine, Histidine, Aspartate

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Chymotrypsin, Trypsin, and Elastase utilize the Ser195-His57-Asp102 catalytic triad for nucleophilic cleavage.

[B] (**INCORRECT**): Cysteine proteases use Cysteine-Histidine pairs.

[C] (**INCORRECT**): Threonine proteases exist in proteasomes, but Ser-His-Asp is the classic triad.

[D] (**INCORRECT**): Tyrosine-Glycine-Proline do not form catalytic triads.

Question 37

Tetrahydrofolate (THF), derived from Folic Acid (Vitamin B9), functions as a coenzyme for:

- A. Acyl transfer reactions
- B. One-carbon unit transfers (methyl, methylene, formyl)
- C. Electron transport chain Complex I
- D. Decarboxylation of pyruvate

Correct Answer: B. One-carbon unit transfers (methyl, methylene, formyl)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): CoA-SH transfers acyl groups.

[B] (**CORRECT**): THF transfers 1-carbon units essential for purine, thymidylate, and methionine synthesis.

[C] (**INCORRECT**): Complex I uses FMN and Fe-S clusters.

[D] (**INCORRECT**): Pyruvate decarboxylation requires TPP.

TOPIC 4: Nomenclature, IUBMB Classification & Mechanisms

Question 38

Under the IUBMB classification system, enzymes belonging to EC Class 1 are designated as:

- A. Transferases
- B. Oxidoreductases
- C. Hydrolases
- D. Lyases

Correct Answer: B. Oxidoreductases

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Transferases are EC Class 2.

[B] (**CORRECT**): EC 1 enzymes are Oxidoreductases (dehydrogenases, oxidases, reductases) catalyzing redox reactions.

[C] (**INCORRECT**): Hydrolases are EC Class 3.

[D] (**INCORRECT**): Lyases are EC Class 4.

Question 39

Kinases, which transfer terminal phosphate groups from ATP to specific substrates, belong to which IUBMB major class?

- A. EC 1: Oxidoreductases
- B. EC 2: Transferases
- C. EC 3: Hydrolases
- D. EC 6: Ligases

Correct Answer: B. EC 2: Transferases

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Oxidoreductases catalyze electron transfers.

[B] (CORRECT): EC 2 Transferases transfer functional groups (phosphate, methyl, amino) between donor and acceptor molecules.

[C] (INCORRECT): Hydrolases cleave bonds by adding water.

[D] (INCORRECT): Ligases join molecules using ATP cleavage.

Question 40

Digestive enzymes such as Trypsin, Pepsin, and Lipase cleave bonds by adding water, classifying them as:

- A. EC 3: Hydrolases
- B. EC 4: Lyases
- C. EC 5: Isomerases
- D. EC 6: Ligases

Correct Answer: A. EC 3: Hydrolases

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Hydrolases (esterases, peptidases, phosphatases) cleave C-O, C-N, or C-P bonds via hydrolysis.

[B] (INCORRECT): Lyases break bonds without hydrolysis or oxidation, often forming double bonds.

[C] (INCORRECT): Isomerases catalyze intramolecular rearrangements.

[D] (INCORRECT): Ligases synthesize bonds coupled to ATP hydrolysis.

Question 41

Fructose-1,6-bisphosphate aldolase cleaves a 6-carbon sugar into two 3-carbon trioses without water. It belongs to:

- A. EC 1: Oxidoreductases
- B. EC 4: Lyases
- C. EC 5: Isomerases
- D. EC 6: Ligases

Correct Answer: B. EC 4: Lyases

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Oxidoreductases require redox cofactors.

[B] (CORRECT): Lyases (aldolases, decarboxylases, dehydratases) cleave C-C, C-O, or C-N bonds non-hydrolytically.

[C] (INCORRECT): Isomerases rearrange structural isomers.

[D] (INCORRECT): Ligases form bonds using ATP.

Question 42

Phosphoglycerate mutase converts 3-phosphoglycerate to 2-phosphoglycerate. Mutases belong to which major class?

- A. EC 2: Transferases
- B. EC 5: Isomerases
- C. EC 3: Hydrolases
- D. EC 4: Lyases

Correct Answer: B. EC 5: Isomerases

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Transferases move groups inter-molecularly between two separate molecules.

[B] (CORRECT): Isomerases (mutases, racemases, epimerases) catalyze intra-molecular functional group shifts within the same molecule.

[C] (INCORRECT): Hydrolases use water.

[D] (INCORRECT): Lyases eliminate groups to form double bonds.

Question 43

Pyruvate Carboxylase and Glutamine Synthetase join two molecules coupled with ATP cleavage, placing them in:

- A. EC 3: Hydrolases
- B. EC 6: Ligases (Synthetases)
- C. EC 4: Lyases
- D. EC 2: Transferases

Correct Answer: B. EC 6: Ligases (Synthetases)

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Hydrolases break bonds using water.

[B] (CORRECT): Ligases (synthetases) join C-C, C-S, C-O, or C-N bonds driven by high-energy nucleoside triphosphate cleavage.

[C] (INCORRECT): Lyases break bonds non-hydrolytically without ATP.

[D] (INCORRECT): Transferases transfer groups without coupled NTP cleavage.

Question 44

What is the key functional nomenclature distinction between a Synthase and a Synthetase?

- A. Synthase requires nucleoside triphosphates (ATP); Synthetase does not
- B. Synthetase requires high-energy phosphate hydrolysis (ATP/GTP); Synthase does not
- C. Synthases are EC 6 enzymes; Synthetases are EC 1 enzymes
- D. Synthetases catalyze cleavage; Synthases catalyze hydrolysis

Correct Answer: B. Synthetase requires high-energy phosphate hydrolysis (ATP/GTP); Synthase does not

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Reversed definition.

[B] (CORRECT): Synthetase (EC 6 Ligase) directly couples bond formation to ATP/GTP hydrolysis (e.g., Glutamine Synthetase); Synthase (EC 4 Lyase) does not (e.g., Citrate Synthase).

[C] (INCORRECT): Synthetases are EC 6; Synthases are typically EC 4.

[D] (INCORRECT): Both catalyze synthesis reactions.

Question 45

What is the key functional nomenclature distinction between a Phosphatase and a Phosphorylase?

- A. Phosphatase uses inorganic phosphate to cleave bonds; Phosphorylase uses water
- B. Phosphatase hydrolytically removes phosphate (Hydrolase); Phosphorylase uses Pi to phosphorolytically cleave bonds (Transferase)
- C. Phosphatase requires ATP; Phosphorylase generates ATP
- D. Both enzymes belong to EC Class 1 Oxidoreductases

Correct Answer: B. Phosphatase hydrolytically removes phosphate (Hydrolase); Phosphorylase uses Pi to phosphorolytically cleave bonds (Transferase)

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Reversed mechanisms.

[B] (CORRECT): Phosphatase (EC 3) uses H₂O to remove phosphate producing free Pi; Phosphorylase (EC 2) uses inorganic phosphate (Pi) to cleave bonds generating phosphorylated products (e.g., Glycogen Phosphorylase).

[C] (INCORRECT): Neither directly consumes ATP.

[D] (INCORRECT): Phosphatases are EC 3; Phosphorylases are EC 2.

Question 46

In general acid-base catalysis, active site amino acid side chains accelerate reactions by acting as proton donors/acceptors. Which amino acid side chain frequently fulfills this role at physiological pH?

- A. Alanine (methyl group)
- B. Histidine (imidazole ring, pKa ~6.0)
- C. Leucine (isobutyl group)
- D. Valine (isopropyl group)

Correct Answer: B. Histidine (imidazole ring, pKa ~6.0)

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Alanine is non-polar and unreactive.

[B] (CORRECT): Histidine's imidazole ring has a pKa near 6.0, allowing it to easily gain or lose protons near physiological pH 7.4.

[C] (INCORRECT): Leucine is hydrophobic.

[D] (INCORRECT): Valine is hydrophobic.

Question 47

Covalent catalysis involves the transient formation of a covalent enzyme-substrate intermediate. Serine proteases utilize which residue to perform nucleophilic attack on the peptide carbonyl?

- A. Serine-195
- B. Phenylalanine-120
- C. Glycine-216
- D. Isoleucine-35

Correct Answer: A. Serine-195

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Serine-195 acts as a potent nucleophile (assisted by His-57 acting as a base) to form an acyl-enzyme intermediate.

[B] (INCORRECT): Phenylalanine forms hydrophobic binding pockets.

[C] (INCORRECT): Glycine provides structural flexibility.

[D] (INCORRECT): Isoleucine is non-polar.

Question 48

Catalysis by approximation (proximity and orientation effect) enhances reaction rates primarily by:

- A. Increasing thermal kinetic motion of substrates randomly
- B. Bringing two substrate molecules into close physical proximity and optimal spatial alignment within the active site
- C. Denaturing surrounding solvent water molecules
- D. Cleaving the enzyme backbone

Correct Answer: B. Bringing two substrate molecules into close physical proximity and optimal spatial alignment within the active site

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Random thermal motion reduces productive collision probability.

[B] (**CORRECT**): Holding reacting groups close together in exact reactive orientation increases effective concentration and rate.

[C] (**INCORRECT**): Active sites exclude bulky water to prevent undesirable side reactions.

[D] (**INCORRECT**): Cleaving enzyme backbone causes denaturation.

Question 49

The oxyanion hole in serine proteases stabilizes catalysis by forming hydrogen bonds with which high-energy intermediate?

- A. Planar ground-state substrate
- B. Tetrahedral intermediate containing a negatively charged oxygen
- C. Fully cleaved uncharged product
- D. Aromatic side chain of the substrate

Correct Answer: B. Tetrahedral intermediate containing a negatively charged oxygen

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Ground-state planar peptide bonds do not fit tightly into the oxyanion hole.

[B] (**CORRECT**): The transition state forms a negative oxygen on the carbonyl (oxyanion), which fits into the oxyanion hole formed by Ser195 and Gly193 backbone NH groups.

[C] (**INCORRECT**): Products release from the site.

[D] (**INCORRECT**): Aromatic side chains fit the hydrophobic S1 binding pocket, not the oxyanion hole.

Question 50

An enzyme EC code 2.7.1.1 designates Hexokinase. The first number (2) specifies that Hexokinase belongs to:

- A. Hydrolases
- B. Transferases
- C. Oxidoreductases
- D. Isomerases

Correct Answer: B. Transferases

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Hydrolases are Class 3.

[B] (**CORRECT**): EC first digit '2' designates Class 2 Transferases (Hexokinase transfers phosphate from ATP to hexoses).

[C] (**INCORRECT**): Oxidoreductases are Class 1.

[D] (**INCORRECT**): Isomerases are Class 5.

TOPIC 5: Factors Affecting Enzyme Activity

Question 51

When substrate is present in excess ($[S] \gg K_m$), doubling the total enzyme concentration ($[Et]$) causes the reaction velocity (v_0) to:

- A. Remain unchanged
- B. Double (increase by 100%)
- C. Decrease by half
- D. Increase exponentially by 4-fold

Correct Answer: B. Double (increase by 100%)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Velocity is directly proportional to enzyme concentration under saturating conditions.

[B] (**CORRECT**): $V_{max} = k_{cat} * [Et]$. Doubling $[Et]$ directly doubles V_{max} and initial velocity v_0 .

[C] (**INCORRECT**): Velocity increases, not decreases.

[D] (**INCORRECT**): The relationship between v_0 and $[Et]$ is strictly linear, not exponential.

Question 52

The Q_{10} temperature coefficient for most human enzyme-catalyzed reactions below their denaturation threshold is approximately:

- A. 1.0
- B. 2.0
- C. 5.0
- D. 10.0

Correct Answer: B. 2.0

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): $Q_{10} = 1.0$ means no temperature dependence.

[B] (**CORRECT**): For most biological enzymes, the reaction rate approximately doubles ($Q_{10} \approx 2.0$) for every 10°C rise in temperature until thermal denaturation begins.

[C] (**INCORRECT**): $Q_{10} = 5.0$ is unrealistically high.

[D] (**INCORRECT**): $Q_{10} = 10.0$ is incorrect.

Question 53

What happens to the initial velocity of a human enzyme-catalyzed reaction when temperature is increased beyond 55°C – 60°C ?

- A. Velocity increases continuously following Arrhenius law
- B. Velocity drops sharply toward zero due to thermal denaturation of protein tertiary structure
- C. K_m decreases to zero
- D. The enzyme converts permanently into a ribozyme

Correct Answer: B. Velocity drops sharply toward zero due to thermal denaturation of protein tertiary structure

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Arrhenius law applies until thermal energy disrupts non-covalent protein folding.

[B] (**CORRECT**): Above $\sim 45^\circ\text{C}$ – 55°C , thermal denaturation unfolds native tertiary structure, destroying active sites and dropping activity to zero.

[C] (**INCORRECT**): Denatured enzymes lose functional K_m altogether.

[D] (**INCORRECT**): Proteins cannot convert into nucleic acid ribozymes.

Question 54

Pepsin exhibits optimal catalytic activity at pH 1.5–2.0, whereas Trypsin operates optimally at pH 7.5–8.0. This difference reflects:

- A. Differences in primary amino acid sequence and physiological digestive microenvironments
- B. Pepsin being a ribozyme and Trypsin being a lipid
- C. Trypsin requiring extreme acidity to protonate active site aspartates
- D. Pepsin operating via zero-order kinetics exclusively

Correct Answer: A. Differences in primary amino acid sequence and physiological digestive microenvironments

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Pepsin (stomach) evolved to operate at acidic pH using catalytic aspartates; Trypsin (small intestine) operates at alkaline pH using Ser-His-Asp triad.

[B] (INCORRECT): Both are protein proteases.

[C] (INCORRECT): Trypsin operates in alkaline duodenal fluid.

[D] (INCORRECT): pH optimum is independent of kinetic order.

Question 55

Altering environmental pH outside an enzyme's optimal range reduces catalytic activity primarily by:

- A. Cleaving peptide bonds along the primary sequence
- B. Altering the protonation/ionization state of active site catalytic amino acid side chains
- C. Converting coenzymes into competitive inhibitors
- D. Increasing total activation energy $\Delta G^\ddagger$ permanently

Correct Answer: B. Altering the protonation/ionization state of active site catalytic amino acid side chains

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Moderate pH shifts do not hydrolyze peptide bonds.

[B] (CORRECT): pH changes alter protonation of essential nucleophilic or general acid/base side chains (e.g., His, Glu, Asp, Cys), disrupting binding and catalysis.

[C] (INCORRECT): Coenzymes are not converted to competitive inhibitors by pH.

[D] (INCORRECT): Reversible pH changes do not alter $\Delta G^\ddagger$ permanently.

Question 56

At very low substrate concentrations ($[S] \ll K_m$), the initial rate of an enzyme reaction (v_0) exhibits:

- A. Zero-order kinetics with respect to substrate
- B. First-order kinetics with respect to substrate
- C. Second-order kinetics with respect to enzyme only
- D. Complete independence from enzyme concentration

Correct Answer: B. First-order kinetics with respect to substrate

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Zero-order kinetics occurs at saturating $[S] \gg K_m$.

[B] (CORRECT): When $[S] \ll K_m$, $v_0 = (V_{\max}/K_m) \cdot [S]$, making initial velocity directly proportional (first-order) to $[S]$.

[C] (INCORRECT): It is first-order with respect to substrate.

[D] (INCORRECT): v_0 remains linearly dependent on enzyme concentration.

Question 57

At saturating substrate concentrations ($[S] \gg K_m$), initial reaction velocity (v_0) approaches V_{max} and exhibits:

- A. First-order kinetics with respect to substrate
- B. Zero-order kinetics with respect to substrate
- C. Exponential kinetics
- D. Substrate-induced competitive inhibition

Correct Answer: B. Zero-order kinetics with respect to substrate

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): First-order kinetics occurs at low $[S] \ll K_m$.

[B] (**CORRECT**): When active sites are fully saturated ($[S] \gg K_m$), adding more substrate produces no further increase in rate ($v_0 \approx V_{max}$, zero-order in $[S]$).

[C] (**INCORRECT**): Hyperbolic saturation curve levels off to a horizontal asymptote.

[D] (**INCORRECT**): Saturating substrate leads to maximal velocity, not competitive inhibition.

Question 58

Why are initial velocity (v_0) measurements taken immediately after mixing enzyme and substrate?

- A. To allow product accumulation to reach equilibrium
- B. To minimize reverse reaction rates and product inhibition
- C. Because enzymes denature within 5 seconds
- D. To measure zero-order enzyme destruction

Correct Answer: B. To minimize reverse reaction rates and product inhibition

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): At equilibrium, net reaction rate is zero.

[B] (**CORRECT**): Measuring v_0 at $t=0$ ensures $[Product] \approx 0$, eliminating reverse reaction ($P \rightarrow S$) and product inhibition.

[C] (**INCORRECT**): Stable enzymes operate for minutes to hours at optimal temperature.

[D] (**INCORRECT**): Enzymes are catalysts and are not destroyed during initial rate measurement.

Question 59

Extreme ionic strength (very high salt concentration) suppresses enzyme activity primarily by:

- A. Screening electrostatic interactions and disrupting ionic bonds stabilizing 3D protein structure
- B. Phosphorylating active site serine residues
- C. Hydrolyzing peptide bonds
- D. Oxidizing coenzymes permanently

Correct Answer: A. Screening electrostatic interactions and disrupting ionic bonds stabilizing 3D protein structure

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): High salt shields charged surface groups, breaking salt bridges and causing protein precipitation ('salting out').

[B] (**INCORRECT**): Salt ions do not phosphorylate residues; kinases perform phosphorylation.

[C] (**INCORRECT**): Salt does not cleave peptide bonds.

[D] (**INCORRECT**): Salt ions do not oxidize coenzymes.

Question 60

The incubation temperature at which an enzyme demonstrates maximum catalytic activity during a fixed assay time is its:

- A. Melting temperature (T_m)
- B. Optimum temperature
- C. Absolute zero
- D. Denaturation point

Correct Answer: B. Optimum temperature

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): T_m is the temperature at which 50% of the protein is denatured.

[B] (**CORRECT**): Optimum temperature balances increased kinetic rate against thermal denaturation (typically $\sim 37^\circ\text{C}$ for human enzymes).

[C] (**INCORRECT**): Absolute zero (-273.15°C) halts all molecular motion.

[D] (**INCORRECT**): Denaturation point is where loss of activity predominates.

Question 61

Accumulation of reaction products over time decreases reaction velocity due to:

- A. Product inhibition and increased reverse reaction rate
- B. Increase in total enzyme concentration
- C. Decrease in K_m
- D. Activation of zymogens

Correct Answer: A. Product inhibition and increased reverse reaction rate

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): As products accumulate, they occupy active sites (product inhibition) and drive the reverse reaction.

[B] (**INCORRECT**): Enzyme concentration remains constant.

[C] (**INCORRECT**): K_m is an intrinsic property and does not decrease.

[D] (**INCORRECT**): Zymogen activation increases, rather than decreases, activity.

Question 62

What effect does increasing substrate concentration have on initial velocity (v_0) when $[S]$ is equal to K_m ?

- A. v_0 equals $0.25 V_{\max}$
- B. v_0 equals $0.50 V_{\max}$
- C. v_0 equals $1.00 V_{\max}$
- D. v_0 equals $2.00 V_{\max}$

Correct Answer: B. v_0 equals $0.50 V_{\max}$

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): $v_0 = 0.25 V_{\max}$ occurs at $[S] = 1/3 K_m$.

[B] (**CORRECT**): By definition, K_m is the substrate concentration at which initial velocity is exactly half of maximal velocity ($v_0 = 0.5 V_{\max}$).

[C] (**INCORRECT**): $v_0 = V_{\max}$ occurs at infinite $[S]$.

[D] (**INCORRECT**): v_0 can never exceed $1.0 V_{\max}$.

TOPIC 6: Michaelis-Menten Kinetics & Lineweaver-Burk Analysis

Question 63

The steady-state assumption proposed by Briggs and Haldane in deriving the Michaelis-Menten equation states that:

- A. Substrate concentration decreases to zero instantly
- B. The concentration of the enzyme-substrate complex [ES] remains constant over time
- C. Maximal velocity V_{max} is reached at zero substrate concentration
- D. The rate of ES formation is zero

Correct Answer: B. The concentration of the enzyme-substrate complex [ES] remains constant over time

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Substrate is present in vast excess during steady state.

[B] (**CORRECT**): Steady-state assumes rate of ES formation equals rate of ES breakdown: $d[ES]/dt = 0$.

[C] (**INCORRECT**): V_{max} requires saturating substrate.

[D] (**INCORRECT**): ES forms and breaks down at equal rates; the rate is not zero.

Question 64

The Michaelis constant (K_m) is an operational parameter defined as:

- A. The enzyme concentration required to reach V_{max}
- B. The substrate concentration at which initial velocity is 50% of V_{max}
- C. The rate constant of peptide bond formation
- D. The reciprocal of maximal velocity ($1/V_{max}$)

Correct Answer: B. The substrate concentration at which initial velocity is 50% of V_{max}

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): K_m is a substrate concentration, not an enzyme concentration.

[B] (**CORRECT**): $K_m = [S]$ when $v_0 = 0.5 V_{max}$. It reflects affinity: lower K_m indicates higher substrate affinity.

[C] (**INCORRECT**): K_m is expressed in concentration units (M, mM, μ M), not s⁻¹.

[D] (**INCORRECT**): $1/V_{max}$ is the y-intercept of a Lineweaver-Burk plot.

Question 65

An enzyme with a VERY LOW K_m for a particular substrate indicates that the enzyme:

- A. Has high affinity for that substrate and reaches half-maximal velocity at low substrate concentration
- B. Has low affinity for that substrate and requires high substrate concentration to reach half-maximal velocity
- C. Is catalytically inactive
- D. Exhibits sigmoidal allosteric kinetics

Correct Answer: A. Has high affinity for that substrate and reaches half-maximal velocity at low substrate concentration

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Low K_m means tight binding / high affinity; small amounts of substrate are sufficient to half-saturate active sites.

[B] (**INCORRECT**): High K_m reflects low affinity (e.g., Glucokinase $K_m \sim 10$ mM vs Hexokinase $K_m \sim 0.05$ mM).

[C] (**INCORRECT**): K_m reflects binding affinity, not catalytic inactivity.

[D] (**INCORRECT**): Michaelis-Menten enzymes show hyperbolic curves; allosteric enzymes show sigmoidal curves.

Question 66

The catalytic turnover number (k_{cat}) represents:

- A. The number of substrate molecules converted to product per enzyme active site per unit time at saturation
- B. The substrate concentration at 50% V_{max}
- C. The energy required to denature the enzyme
- D. The ratio of enzyme to coenzyme

Correct Answer: A. The number of substrate molecules converted to product per enzyme active site per unit time at saturation

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): $k_{cat} = V_{max} / [E_t]$, expressed in s⁻¹, measures maximal catalytic conversion frequency per active site.

[B] (**INCORRECT**): K_m defines 50% V_{max} substrate concentration.

[C] (**INCORRECT**): Denaturation energy is activation energy of unfolding.

[D] (**INCORRECT**): k_{cat} is a microscopic rate constant.

Question 67

Which parameter serves as the ultimate measure of catalytic efficiency for comparing an enzyme's preference among competing substrates?

- A. K_m alone
- B. V_{max} alone
- C. k_{cat} / K_m (specificity constant)
- D. Half-life (t_{1/2})

Correct Answer: C. k_{cat} / K_m (specificity constant)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): K_m measures binding affinity only, ignoring catalytic conversion rate.

[B] (**INCORRECT**): V_{max} depends on enzyme concentration.

[C] (**CORRECT**): k_{cat}/K_m combines binding affinity (1/K_m) and catalytic rate (k_{cat}); upper physical limit is diffusion (~10⁸–10⁹ M⁻¹s⁻¹).

[D] (**INCORRECT**): Half-life measures protein degradation.

Question 68

On a Lineweaver-Burk double-reciprocal plot (1/v₀ vs 1/[S]), the X-intercept represents:

- A. 1 / V_{max}
- B. -1 / K_m
- C. K_m / V_{max}
- D. k_{cat}

Correct Answer: B. -1 / K_m

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): 1/V_{max} is the Y-intercept.

[B] (**CORRECT**): Setting 1/v₀ = 0 in the Lineweaver-Burk equation yields 1/[S] = -1/K_m.

[C] (**INCORRECT**): K_m/V_{max} is the slope of the line.

[D] (**INCORRECT**): k_{cat} is calculated from V_{max}/[E_t].

Question 69

On a Lineweaver-Burk double-reciprocal plot ($1/v_0$ vs $1/[S]$), the Y-intercept represents:

- A. $1 / V_{\max}$
- B. $-1 / K_m$
- C. $K_m / V_{\max}$
- D. $V_{\max} / K_m$

Correct Answer: A. $1 / V_{\max}$

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Setting $1/[S] = 0$ (infinite substrate) yields $1/v_0 = 1/V_{\max}$ at the Y-intercept.

[B] (**INCORRECT**): $-1/K_m$ is the X-intercept.

[C] (**INCORRECT**): $K_m/V_{\max}$ is the slope.

[D] (**INCORRECT**): $V_{\max}/K_m$ is the inverse slope.

Question 70

On a Lineweaver-Burk double-reciprocal plot, the SLOPE of the straight line equals:

- A. $K_m / V_{\max}$
- B. $V_{\max} / K_m$
- C. $-1 / K_m$
- D. $1 / V_{\max}$

Correct Answer: A. $K_m / V_{\max}$

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): The linearized equation is $(1/v_0) = (K_m/V_{\max}) \cdot (1/[S]) + (1/V_{\max})$. The slope is $K_m/V_{\max}$.

[B] (**INCORRECT**): $V_{\max}/K_m$ is the inverse slope.

[C] (**INCORRECT**): $-1/K_m$ is the X-intercept.

[D] (**INCORRECT**): $1/V_{\max}$ is the Y-intercept.

Question 71

Hexokinase I has a K_m for glucose of ~ 0.05 mM, whereas liver Glucokinase (Hexokinase IV) has a K_m of ~ 10 mM. What is the physiological consequence?

- A. Hexokinase I operates near $V_{\max}$ at normal fasting blood glucose (~ 5 mM), while Glucokinase rate increases proportionally after carbohydrate meals
- B. Glucokinase is fully saturated during hypoglycemia
- C. Hexokinase I cannot phosphorylate glucose
- D. Glucokinase has 200-fold higher affinity for glucose than Hexokinase I

Correct Answer: A. Hexokinase I operates near $V_{\max}$ at normal fasting blood glucose (~ 5 mM), while Glucokinase rate increases proportionally after carbohydrate meals

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Low K_m (~ 0.05 mM) allows brain Hexokinase to phosphorylate glucose even during fasting. High K_m (~ 10 mM) allows liver Glucokinase to clear blood glucose efficiently postprandially.

[B] (**INCORRECT**): Glucokinase is inactive during hypoglycemia due to high K_m .

[C] (**INCORRECT**): Hexokinase I rapidly phosphorylates glucose.

[D] (**INCORRECT**): Glucokinase has 200-fold lower affinity (higher K_m).

Question 72

Given $V_{\max} = 100 \mu\text{mol/min}$ and $K_m = 2 \text{ mM}$, calculate initial velocity (v_0) when substrate concentration $[S] = 2 \text{ mM}$:

- A. $25 \mu\text{mol/min}$
- B. $50 \mu\text{mol/min}$
- C. $75 \mu\text{mol/min}$
- D. $100 \mu\text{mol/min}$

Correct Answer: B. $50 \mu\text{mol/min}$

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): $25 \mu\text{mol/min}$ occurs at $[S] = 0.67 \text{ mM}$.

[B] (CORRECT): When $[S] = K_m$, $v_0 = 0.5 * V_{\max} = 0.5 * 100 = 50 \mu\text{mol/min}$.

[C] (INCORRECT): $75 \mu\text{mol/min}$ occurs at $[S] = 6 \text{ mM}$.

[D] (INCORRECT): $100 \mu\text{mol/min}$ is theoretical $V_{\max}$ at infinite $[S]$.

Question 73

Using the Michaelis-Menten equation, if $[S] = 3 K_m$, what fraction of $V_{\max}$ is achieved?

- A. $0.25 V_{\max}$
- B. $0.50 V_{\max}$
- C. $0.75 V_{\max}$
- D. $0.90 V_{\max}$

Correct Answer: C. $0.75 V_{\max}$

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): $0.25 V_{\max}$ occurs at $[S] = 1/3 K_m$.

[B] (INCORRECT): $0.50 V_{\max}$ occurs at $[S] = K_m$.

[C] (CORRECT): $v_0 = V_{\max} * (3 K_m) / (K_m + 3 K_m) = 3/4 V_{\max} = 0.75 V_{\max}$.

[D] (INCORRECT): $0.90 V_{\max}$ occurs at $[S] = 9 K_m$.

Question 74

A reaction has a $V_{\max}$ of 200 nmol/s . If total enzyme concentration $[E_t]$ is 2 nmol , what is k_{cat} ?

- A. 10 s^{-1}
- B. 100 s^{-1}
- C. 400 s^{-1}
- D. 1000 s^{-1}

Correct Answer: B. 100 s^{-1}

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): 10 s^{-1} is incorrect.

[B] (CORRECT): $k_{\text{cat}} = V_{\max} / [E_t] = (200 \text{ nmol/s}) / (2 \text{ nmol}) = 100 \text{ s}^{-1}$.

[C] (INCORRECT): 400 s^{-1} is incorrect.

[D] (INCORRECT): 1000 s^{-1} is incorrect.

Question 75

Enzymes that obey classic Michaelis-Menten kinetics display what shape curve when v_0 is plotted against $[S]$?

- A. Sigmoidal (S-shaped)
- B. Hyperbolic
- C. Linear positive slope
- D. Bell-shaped

Correct Answer: B. Hyperbolic

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Sigmoidal curves are characteristic of allosteric enzymes (e.g., PFK-1, ATCase).

[B] (**CORRECT**): Classic Michaelis-Menten enzymes yield a rectangular hyperbolic v_0 vs $[S]$ curve.

[C] (**INCORRECT**): Linear curves reflect first-order kinetics at low $[S]$ before saturation.

[D] (**INCORRECT**): Bell-shaped curves reflect v_0 vs pH plots.

TOPIC 7: Regulation of Enzyme Activity & Inhibitors

Question 76

A reversible Competitive Inhibitor binds to which site on the enzyme?

- A. An allosteric regulatory site exclusively
- B. The free enzyme active site, competing directly with substrate
- C. The enzyme-substrate complex $[ES]$ exclusively
- D. The carbohydrate side chains

Correct Answer: B. The free enzyme active site, competing directly with substrate

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Allosteric inhibitors bind distant regulatory sites.

[B] (**CORRECT**): Competitive inhibitors structurally resemble the substrate and bind reversibly to the free active site.

[C] (**INCORRECT**): Uncompetitive inhibitors bind the ES complex.

[D] (**INCORRECT**): Carbohydrate side chains are not active site targets.

Question 77

How do kinetic parameters K_m and V_{max} change in the presence of a REVERSIBLE COMPETITIVE INHIBITOR?

- A. Apparent K_m increases; V_{max} remains unchanged
- B. K_m remains unchanged; V_{max} decreases
- C. Both K_m and V_{max} decrease
- D. Both K_m and V_{max} increase

Correct Answer: A. Apparent K_m increases; V_{max} remains unchanged

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Competitive inhibitors increase apparent K_m (requires more substrate to reach 50% V_{max}). At saturating $[S]$, inhibitor is displaced, leaving V_{max} unchanged.

[B] (**INCORRECT**): Unchanged K_m with decreased V_{max} characterizes noncompetitive inhibition.

[C] (**INCORRECT**): Decreased K_m and V_{max} characterizes uncompetitive inhibition.

[D] (**INCORRECT**): Increased K_m and V_{max} is not standard.

Question 78

Statins (e.g., Atorvastatin) lower plasma cholesterol by acting as competitive inhibitors of which rate-limiting enzyme?

- A. Lipoprotein Lipase
- B. HMG-CoA Reductase
- C. ACAT
- D. LCAT

Correct Answer: B. HMG-CoA Reductase

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Lipoprotein lipase clears chylomicron/VLDL triglycerides.

[B] (**CORRECT**): Statins structurally resemble HMG-CoA and competitively inhibit HMG-CoA reductase, blocking mevalonate synthesis.

[C] (**INCORRECT**): ACAT esterifies intracellular cholesterol.

[D] (**INCORRECT**): LCAT esterifies cholesterol in HDL.

Question 79

How do kinetic parameters K_m and V_{max} change in the presence of a REVERSIBLE NONCOMPETITIVE INHIBITOR?

- A. Apparent K_m increases; V_{max} remains unchanged
- B. K_m remains unchanged; V_{max} decreases
- C. Both K_m and V_{max} decrease
- D. K_m decreases; V_{max} increases

Correct Answer: B. K_m remains unchanged; V_{max} decreases

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Increased K_m with unchanged V_{max} is competitive.

[B] (**CORRECT**): Noncompetitive inhibitors bind E or ES at an allosteric site equally well. Substrate binding (K_m) is unaffected, but catalytic rate (V_{max}) decreases.

[C] (**INCORRECT**): Both decreasing is uncompetitive.

[D] (**INCORRECT**): Incorrect.

Question 80

Heavy metals such as Lead (Pb^{2+}) and Mercury (Hg^{2+}) inhibit enzymes by binding to sulfhydryl (-SH) cysteine groups allosterically. They act as:

- A. Competitive inhibitors
- B. Noncompetitive / Irreversible inhibitors
- C. Coenzymes
- D. Allosteric activators

Correct Answer: B. Noncompetitive / Irreversible inhibitors

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): They do not compete with substrate for active site binding pockets.

[B] (**CORRECT**): Heavy metals bind essential cysteine -SH groups, disrupting active sites noncompetitively/irreversibly and lowering functional V_{max} .

[C] (**INCORRECT**): Coenzymes assist catalysis.

[D] (**INCORRECT**): Activators increase rate.

Question 81

Aspirin (Acetylsalicylic acid) irreversibly inhibits Cyclooxygenase (COX-1 and COX-2) by performing which covalent modification?

- A. Phosphorylation of tyrosine residues
- B. Acetylation of an active site serine residue
- C. Ubiquitination of lysine residues
- D. Glycosylation of asparagine residues

Correct Answer: B. Acetylation of an active site serine residue

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Kinases phosphorylate tyrosine.

[B] (CORRECT): Aspirin transfers its acetyl group to Ser-530 (COX-1) or Ser-516 (COX-2), irreversibly blocking arachidonic acid entry and thromboxane/prostaglandin synthesis.

[C] (INCORRECT): Ubiquitination targets degradation.

[D] (INCORRECT): Glycosylation occurs in Golgi.

Question 82

Allopurinol acts as a suicide inhibitor (mechanism-based inhibitor) of Xanthine Oxidase. It is metabolized to Alloxanthine, which:

- A. Binds tightly to the active site molybdenum-iron complex, halting uric acid production
- B. Phosphorylates glycogen phosphorylase
- C. Hydrolyzes ATP into cAMP
- D. Cleaves trypsinogen

Correct Answer: A. Binds tightly to the active site molybdenum-iron complex, halting uric acid production

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Xanthine oxidase oxidizes Allopurinol into Alloxanthine, which binds irreversibly to Mo-Fe at the active site, treating gout.

[B] (INCORRECT): Phosphorylase kinase phosphorylates glycogen phosphorylase.

[C] (INCORRECT): Adenylyl cyclase forms cAMP.

[D] (INCORRECT): Enteropeptidase cleaves trypsinogen.

Question 83

Allosteric enzymes (e.g., Phosphofructokinase-1) display sigmoidal v_0 vs $[S]$ curves. Heterotropic positive allosteric effectors:

- A. Shift the sigmoidal curve to the right (increase $K_{0.5}$)
- B. Shift the sigmoidal curve to the left (decrease $K_{0.5}$), stabilizing the active R (relaxed) state
- C. Destroy quaternary structure permanently
- D. Function as competitive inhibitors

Correct Answer: B. Shift the sigmoidal curve to the left (decrease $K_{0.5}$), stabilizing the active R (relaxed) state

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Rightward shifts (increased $K_{0.5}$) are caused by allosteric inhibitors stabilizing the T (tense) state.

[B] (CORRECT): Positive effectors (e.g., AMP, F-2,6-BP for PFK-1) bind regulatory sites, stabilizing the R-state, lowering $K_{0.5}$, and shifting curves leftward.

[C] (INCORRECT): Allosteric regulation is non-destructive and reversible.

[D] (INCORRECT): Allosteric effectors do not compete at active sites.

Question 84

Organophosphate nerve agents (e.g., Sarin, Diisopropylfluorophosphate) irreversibly inhibit Acetylcholinesterase by covalently modifying:

- A. Active site serine hydroxyl group
- B. C-terminal lysine carboxyl group
- C. N-terminal alanine amino group
- D. Histidine imidazole ring

Correct Answer: A. Active site serine hydroxyl group

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Organophosphates phosphorylate the active site Ser-200 of Acetylcholinesterase, causing acetylcholine accumulation, cholinergic crisis, and paralysis.

[B] (INCORRECT): Lysine is not the target.

[C] (INCORRECT): N-terminus is not involved.

[D] (INCORRECT): His is part of triad, but Ser is covalently modified.

Question 85

Reversible covalent modification of enzymes via phosphorylation is catalyzed by Protein Kinases, which transfer phosphate from ATP to:

- A. Hydroxyl groups of Serine, Threonine, or Tyrosine residues
- B. Carboxyl groups of Aspartate and Glutamate
- C. Amino groups of Lysine and Arginine
- D. Thiol groups of Cysteine

Correct Answer: A. Hydroxyl groups of Serine, Threonine, or Tyrosine residues

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Protein Kinases transfer γ -phosphate of ATP to hydroxyl (-OH) side chains of Ser, Thr, or Tyr residues.

[B] (INCORRECT): Asp/Glu carboxyls are not standard eukaryotic kinase targets.

[C] (INCORRECT): Lysine phosphorylation is rare and non-canonical.

[D] (INCORRECT): Cysteine thiols undergo oxidation/S-nitrosylation, not standard kinase phosphorylation.

Question 86

Phosphorylation of Glycogen Phosphorylase and Glycogen Synthase by Protein Kinase A in response to Glucagon results in:

- A. Inactivation of Phosphorylase and Activation of Synthase
- B. Activation of Glycogen Phosphorylase (b $\rightarrow$ a) and Inactivation of Glycogen Synthase (a $\rightarrow$ b)
- C. Activation of both enzymes simultaneously
- D. Inactivation of both enzymes simultaneously

Correct Answer: B. Activation of Glycogen Phosphorylase (b $\rightarrow$ a) and Inactivation of Glycogen Synthase (a $\rightarrow$ b)

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Reversed regulation.

[B] (CORRECT): Phosphorylation coordinately activates glycogen breakdown (Phosphorylase a active) and inhibits glycogen synthesis (Synthase b inactive) to raise blood glucose.

[C] (INCORRECT): Reciprocal regulation prevents wasteful futile cycles.

[D] (INCORRECT): Simultaneous inactivation would halt carbohydrate metabolism.

Question 87

Zymogen activation (e.g., Trypsinogen to Trypsin by Enteropeptidase) is physiologically unique because it is:

- A. Reversibly regulated by ATP concentration
- B. An irreversible covalent specific proteolytic cleavage occurring outside the synthesis site
- C. Catalyzed exclusively by ribozymes
- D. Mediated by allosteric feedback inhibition

Correct Answer: B. An irreversible covalent specific proteolytic cleavage occurring outside the synthesis site

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Zymogen activation is irreversible, not ATP-dependent.

[B] (CORRECT): Proteolytic cleavage of a specific peptide bond activates zymogens irreversibly in the intestinal lumen, protecting the pancreas from autodigestion.

[C] (INCORRECT): Enteropeptidase is a protein enzyme.

[D] (INCORRECT): It is proteolytic cleavage, not allosteric regulation.

Question 88

Enzyme Induction and Repression regulate metabolic pathways by:

- A. Altering substrate binding affinity (K_m) within milliseconds
- B. Controlling gene transcription and rate of de novo enzyme protein synthesis over hours to days
- C. Cleaving peptide zymogens instantly
- D. Phosphorylating active site histidine residues

Correct Answer: B. Controlling gene transcription and rate of de novo enzyme protein synthesis over hours to days

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Allosteric regulation occurs in milliseconds.

[B] (CORRECT): Induction (e.g., Insulin inducing glucokinase) and Repression alter total enzyme concentration $[E_t]$ by regulating mRNA transcription over hours/days.

[C] (INCORRECT): Zymogen activation is rapid proteolysis.

[D] (INCORRECT): Covalent phosphorylation takes seconds to minutes.

TOPIC 8: Diagnostic & Clinical Utility of Enzymes & Isoenzymes

Question 89

How do functional plasma enzymes differ from non-functional plasma enzymes?

- A. Functional plasma enzymes perform physiological roles in blood (e.g., blood clotting factors); non-functional enzymes leak from damaged tissue cells into blood
- B. Functional plasma enzymes are ribozymes; non-functional enzymes are lipids
- C. Non-functional plasma enzymes have infinite V_{max} in blood
- D. Functional plasma enzymes are cleared within 2 seconds by kidneys

Correct Answer: A. Functional plasma enzymes perform physiological roles in blood (e.g., blood clotting factors); non-functional enzymes leak from damaged tissue cells into blood

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Functional plasma enzymes (thrombin, lipoprotein lipase) act in plasma. Non-functional enzymes (ALT, AST, CK) function intracellularly and appear in blood only after tissue cell damage.

[B] (INCORRECT): Both are protein enzymes.

[C] (INCORRECT): Non-functional enzymes operate under standard kinetics.

[D] (INCORRECT): Plasma half-lives range from hours to days.

Question 90

Isoenzymes (Isozymes) are defined as physically distinct forms of an enzyme that:

- A. Catalyze completely different chemical reactions
- B. Catalyze the same chemical reaction but differ in amino acid sequence, kinetic properties (K_m , V_{max}), and electrophoretic mobility
- C. Are composed of nucleic acids
- D. Are active only after heat denaturation

Correct Answer: B. Catalyze the same chemical reaction but differ in amino acid sequence, kinetic properties (K_m , V_{max}), and electrophoretic mobility

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Isozymes catalyze the exact same reaction.

[B] (CORRECT): Isozymes are encoded by different genes or splice variants, displaying different tissue distribution, K_m , regulatory controls, and net charge.

[C] (INCORRECT): Isozymes are proteins.

[D] (INCORRECT): Denaturation destroys activity.

Question 91

Creatine Kinase (CK) exists as three dimeric isoenzymes. Which isoform is predominantly found in cardiac muscle and serves as a diagnostic marker for myocardial infarction?

- A. CK-1 (BB)
- B. CK-2 (MB)
- C. CK-3 (MM)
- D. CK-4 (mt)

Correct Answer: B. CK-2 (MB)

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): CK-1 (BB) is localized in brain tissue.

[B] (CORRECT): CK-2 (MB) constitutes ~20–30% of cardiac CK (vs <1% in skeletal muscle), making elevated CK-MB specific for acute myocardial injury.

[C] (INCORRECT): CK-3 (MM) is the predominant isoform in skeletal muscle (~98%).

[D] (INCORRECT): Mitochondrial CK is non-sarcolemmal.

Question 92

Following an acute myocardial infarction (AMI), serum CK-MB levels typically begin to rise within 4–8 hours, peak at 12–24 hours, and return to baseline after:

- A. 12 hours
- B. 48–72 hours (2–3 days)
- C. 14–21 days
- D. 6 months

Correct Answer: B. 48–72 hours (2–3 days)

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): CK-MB peaks at 12–24 hours.

[B] (CORRECT): CK-MB returns to baseline in 48–72 hours, making it valuable for detecting early re-infarction.

[C] (INCORRECT): Cardiac troponins (cTnI, cTnT) remain elevated for 7–14 days.

[D] (INCORRECT): 6 months is far beyond enzyme clearance.

Question 93

In healthy serum, Lactate Dehydrogenase isoenzyme LDH-2 (H3M) exceeds LDH-1 (H4). A 'flipped LDH pattern' (LDH-1 > LDH-2) is characteristic of:

- A. Acute Myocardial Infarction or hemolytic anemia
- B. Acute Pancreatitis
- C. Prostate Cancer
- D. Obstructive Jaundice

Correct Answer: A. Acute Myocardial Infarction or hemolytic anemia

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): Heart muscle and RBCs are rich in LDH-1. Necrosis releases LDH-1, causing LDH-1 level to exceed LDH-2 ('flipped LDH').

[B] (**INCORRECT**): Pancreatitis elevates Amylase and Lipase.

[C] (**INCORRECT**): Prostate cancer elevates PSA and Acid Phosphatase.

[D] (**INCORRECT**): Jaundice elevates ALP and GGT.

Question 94

Serum Alanine Aminotransferase (ALT) is a more specific clinical biomarker for hepatocellular damage than Aspartate Aminotransferase (AST) because:

- A. ALT is present in high concentrations almost exclusively in liver cytosol, whereas AST is abundant in heart, skeletal muscle, and kidneys
- B. ALT is a ribozyme
- C. AST has zero Km for alanine
- D. ALT is cleared instantly by lungs

Correct Answer: A. ALT is present in high concentrations almost exclusively in liver cytosol, whereas AST is abundant in heart, skeletal muscle, and kidneys

Detailed Choice-by-Choice Rationale:

[A] (**CORRECT**): ALT is liver-specific. AST is distributed across liver, cardiac, muscle, and renal tissues; thus elevated ALT specifically pinpoints liver necrosis.

[B] (**INCORRECT**): Both are protein transaminases.

[C] (**INCORRECT**): AST uses aspartate/oxaloacetate.

[D] (**INCORRECT**): Both are cleared by reticuloendothelial uptake.

Question 95

Elevated serum Alkaline Phosphatase (ALP) accompanied by elevated Gamma-Glutamyl Transferase (GGT) indicates disease of the:

- A. Skeletal bone system (Paget's disease)
- B. Hepatobiliary system (bile duct obstruction / cholestasis)
- C. Brain cortex
- D. Skeletal muscle

Correct Answer: B. Hepatobiliary system (bile duct obstruction / cholestasis)

Detailed Choice-by-Choice Rationale:

[A] (**INCORRECT**): Bone diseases elevate ALP alone; GGT is normal in bone disorders.

[B] (**CORRECT**): Concomitant elevation of ALP and GGT confirms hepatobiliary origin (biliary obstruction, cholestasis) rather than bone pathology.

[C] (**INCORRECT**): Brain injury elevates CK-BB.

[D] (**INCORRECT**): Muscle damage elevates CK-MM and AST.

Question 96

Serum Gamma-Glutamyl Transferase (GGT) is a sensitive biomarker for detecting recent consumption of:

- A. Alcohol (Ethanol) and microsomal enzyme-inducing drugs (e.g., Phenytoin)
- B. Vitamin C megadoses
- C. High carbohydrate meals
- D. Saturated fat

Correct Answer: A. Alcohol (Ethanol) and microsomal enzyme-inducing drugs (e.g., Phenytoin)

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Ethanol induces hepatic microsomal GGT expression, making elevated serum GGT a key marker for heavy alcohol ingestion.

[B] (INCORRECT): Vitamin C does not induce GGT.

[C] (INCORRECT): Carbohydrates do not induce GGT.

[D] (INCORRECT): Fats elevate chylomicrons.

Question 97

Marked elevation of serum Lipase and Amylase (3-fold or greater above upper limit of normal) confirms a diagnosis of:

- A. Acute Pancreatitis
- B. Acute Hepatitis A
- C. Myocardial Infarction
- D. Renal Failure

Correct Answer: A. Acute Pancreatitis

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): Pancreatic acinar damage releases Amylase and Lipase into blood. Lipase is more specific and remains elevated longer.

[B] (INCORRECT): Hepatitis elevates ALT and AST.

[C] (INCORRECT): MI elevates Troponins and CK-MB.

[D] (INCORRECT): Renal failure elevates BUN and Creatinine.

Question 98

Prostate-Specific Antigen (PSA) and Tartrate-Inhibited Acid Phosphatase (ACP) are clinical tumor markers used for screening:

- A. Prostatic adenocarcinoma
- B. Hepatic hemangioma
- C. Small cell lung cancer
- D. Colorectal carcinoma

Correct Answer: A. Prostatic adenocarcinoma

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): PSA (a serine protease) and prostatic acid phosphatase are elevated in prostate cancer and benign prostatic hyperplasia.

[B] (INCORRECT): Liver lesions monitor AFP.

[C] (INCORRECT): Lung cancer monitors NSE.

[D] (INCORRECT): Colorectal cancer monitors CEA.

Question 99

Glucose-6-Phosphate Dehydrogenase (G6PD) deficiency causes episodic hemolytic anemia upon exposure to oxidative drugs (e.g., Primaquine, Sulfa drugs) because erythrocyte G6PD is required to generate:

- A. NADP⁺ for glycolysis
- B. NADPH to maintain reduced Glutathione (GSH) for H₂O₂ detoxification
- C. ATP for Na⁺/K⁺ ATPase
- D. FAD for succinate oxidation

Correct Answer: B. NADPH to maintain reduced Glutathione (GSH) for H₂O₂ detoxification

Detailed Choice-by-Choice Rationale:

[A] (INCORRECT): Glycolysis generates NADH, not NADPH.

[B] (CORRECT): RBCs rely on G6PD in the pentose phosphate pathway as their sole source of NADPH to regenerate reduced glutathione (GSH) via Glutathione Reductase, detoxifying reactive oxygen species.

[C] (INCORRECT): Glycolysis yields ATP in RBCs.

[D] (INCORRECT): RBCs lack mitochondria and FAD-dependent TCA cycle enzymes.

Question 100

Recombinant Tissue Plasminogen Activator (tPA, Alteplase) is administered therapeutically during acute ischemic stroke or MI because it activates:

- A. Plasminogen to Plasmin, which degrades fibrin blood clots
- B. Prothrombin to Thrombin, accelerating clot formation
- C. Fibrinogen to Fibrin polymers
- D. Trypsinogen to Trypsin in circulation

Correct Answer: A. Plasminogen to Plasmin, which degrades fibrin blood clots

Detailed Choice-by-Choice Rationale:

[A] (CORRECT): tPA is a fibrin-specific serine protease that converts plasminogen to active plasmin, dissolving intravascular thrombi (thrombolysis).

[B] (INCORRECT): Prothrombin conversion promotes clotting, opposing thrombolysis.

[C] (INCORRECT): Fibrin formation creates clots.

[D] (INCORRECT): Trypsinogen is a pancreatic digestive zymogen.